14.2

Let C be a smooth plane curve: $x = x(t)$, $y = y(t)$, $a \le t \le b$

such that $r'(t) = (x'(t), y'(t)) \neq 0$

$$\int_C f(x,y)\, ds = \lim_{\|P\| \to 0} \sum_{i=1}^{n} f(\bar{x}_i, \bar{y}_i)\, \Delta s_i$$

where $a = t_0 < t_1 < t_2 \qquad < t_n = b$

$\bar{x}_i = x(\bar{t}_i)$ $\quad t_i \le \bar{t}_i \le t_{i+1}$

$\bar{y}_i = y(\bar{t}_i)$

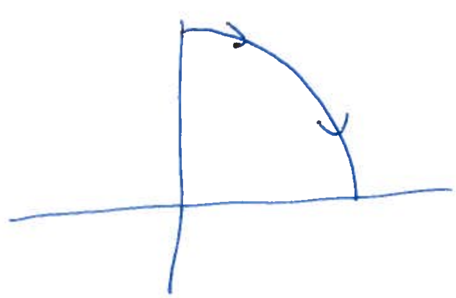
Δs_i = length of arc from $P_i = (x(t_i), y(t_i))$ to P_{i+1}.

$\Delta s_i \approx \|v\| \cdot \Delta t_i \qquad \Delta t_i = t_{i+1} - t_i$

$$\boxed{\int_C f(x,y)\, ds = \int_a^b f(x(t), y(t)) \sqrt{x'(t)^2 + y'(t)^2}\, dt}$$

It does not depend on the choice of the parametrization.

Eg Compute $\int_C y x^2\, ds$ where C = portion of circle of radius 1 in first quadrant, clockwise



$x(t) = \sin t \qquad y(t) = \cos t$

$x'(t) = \cos t \qquad y'(t) = -\sin t$

$0 \le t \le \pi/2$

$$\int_0^{\pi/2} \cos t \sin^2 t \sqrt{\cos^2 t + \sin^2 t}\, dt \qquad u = \sin t \quad du = \cos t\, dt$$

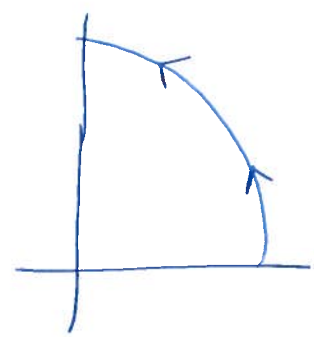
$$= \int u^2\, du = \frac{1}{3}u^3 \Big| = \frac{1}{3}\sin^3 t \Big|_0^{\pi/2} = \frac{1}{3}[0 - 1] = -\frac{1}{3}$$

Eg $x(t) = \sin kt \quad y(t) = \cos kt \qquad 0 \le t \le \frac{\pi}{2k}$

$x'(t) = k \cos kt \quad y'(t) = -k \sin kt$

$$\int_0^{\pi/2k} \cos kt \sin^2 kt \cdot k\, dt \qquad u = \sin kt \quad du = k \cos kt\, dt$$

$$= \int u^2\, du = \frac{u^3}{3}\Big| = \frac{\sin^3 kt}{3}\Big|_0^{\pi/2k} = -\frac{1}{3}$$



Eg counterclockwise $x(t) = \cos t$

$y(t) = \sin t$

$0 \le t \le \pi/2$

$$\int_0^{\pi/2} \sin t \cos^2 t \sqrt{\sin^2 t + \cos^2 t}\, dt = \int_0^{\pi/2} \sin t \cos^2 t\, dt \qquad \cos t = u \quad du = -\sin t\, dt$$

$$= -\int u^2\, du = -\frac{u^3}{3}\Big| = -\frac{\cos^3 t}{3}\Big|_0^{\pi/2} = -\left[0 - \frac{1}{3}\right] = \frac{1}{3}$$

Eg $\int_C (x^3 + y)\, ds$ $\quad C$ is the curve $y = \frac{x^3}{27}$ between $(0,0)$ & $(3,1)$

$x(t) = 3t \qquad 0 \le t \le 1$

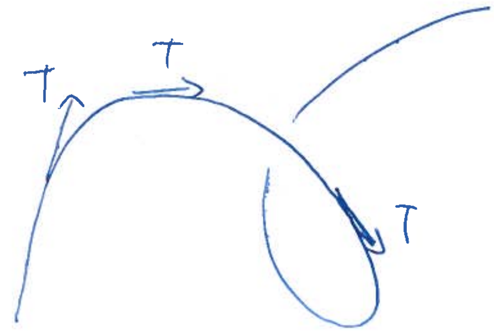
$y(t) = t^3$

$$\int_0^1 (27t^3 + t^3) \cdot \sqrt{9 + 9t^4}\, dt$$

$$= \int_0^1 28t^3 \cdot 3\sqrt{1+t^4}\, dt \qquad u = 1 + t^4 \quad du = 4t^3\, dt$$

$$= \int 7 \cdot 3 \sqrt{u}\, du = 21 \cdot \frac{2}{3} u^{3/2} \Big| = 14 \cdot (1+t^4)^{3/2} \Big|_0^1 = 14\left(2^{3/2} - 1\right)$$

$$\text{Work} = \int_C \underbrace{F\cdot T}_{\substack{\uparrow \\ \text{parallel component of } F}} \, ds$$



$$W = \int_C F\cdot T\,ds = \int_C F\cdot dr$$

(Recall that $T = \dfrac{dr/dt}{dt/ds} = dr/ds$)

If $F = (M, N, P)$ & $dr = (dx, dy, dz)$ then

$$W = \int_C M\,dx + N\,dy + P\,dz$$

Ex Find the work done by $F = (x^2 - y^2)\,i + 2xy\,j$ along the curve $y^2 = x^3$ between $(0,0)$ & $(1,1)$

$x = t^2 \quad y = t^3 \qquad 0 \le t \le 1 \qquad dx = 2t\,dt \quad dy = 3t^2\,dt$

$$\int_0^1 (t^4 - t^6)\cdot 2t\,dt + 2t^5\cdot 3t^2\,dt = \left[\tfrac{1}{3}t^6 - \tfrac{1}{4}t^8 + \tfrac{3t^8}{4}\right]_0^1 = \tfrac{1}{3} + \tfrac{1}{2} = \tfrac{5}{6}$$

Ex Find the work done by $F(x,y) = (e^x, -e^{-y})$ along the curve $x = 3\ln t$, $y = \ln 2t$ $\quad 1 \le t \le 5$

$dx = \frac{3}{t}\,dt$, $dy = \frac{2\,dt}{2t} = \frac{dt}{t}$

$$\int_1^5 e^{3\ln t}\,\frac{3}{t}\,dt - e^{-\ln 2t}\,\frac{dt}{t} = \int_1^5 \frac{3t^3}{t} - \frac{1}{2t}\cdot\frac{1}{t}\,dt = \left[t^3 + \frac{1}{2t}\right]_1^5$$

$$= 125 + \tfrac{1}{10} - 1 - \tfrac{1}{2}$$

Ex Evaluate the following integrals.

a) $\int_C (\sin x + \cos y)\, ds$ $\quad$ C is the line segment from $(0,0)$ to $(\pi, 2\pi)$

b) $\int_C (2x + 9z)\, ds$ $\quad$ $C = \{x = t,\ y = t^2,\ z = t^3,\ 0 \le t \le 1\}$

c) $\int (x+y-z)\, dx + x\, dy - yz\, dz$ $\quad$ C the line segment from $(1,2,1)$ to $(2,1,0)$

Ex Find the work done by the force F in moving a particle along the curve C.

a) $F(x,y) = (x+y)\,i + (x-y)\,j$ $\quad$ $C = \{x = a\cos t,\ y = b\sin t,\ 0 \le t \le \pi/2\}$

b) $F(x,y,z) = (2x-y)\,i + 2z\,j + (y-z)\,k$

C the line from $(0,0,0)$ to $(1,1,1)$

c) $F(x,y,z) = (y, z, x)$ $\quad$ $C = \{x = t,\ y = t^2,\ z = t^3,\ 0 \le t \le 2\}$